Mid-Term Stochastic Scheduling of a Price-Maker Hydro Producer With Pumped Storage

Costas G. Baslis, *Member, IEEE*, and Anastasios G. Bakirtzis, *Senior Member, IEEE*

***Abstract*—A yearly stochastic self-scheduling model for a price-maker hydro producer is presented. In the short term, the producer aims at maximizing profits in the day-ahead market. Residual demand curves model the producer's interaction with his competitors and the load demand. A modification of the residual demand curve is proposed to enable the definition of optimal pumped-hydro bids. Multistage stochastic programming provides hedging against the mid-term uncertainty of inflows, load demand, and competitors' offers. The proposed method models the impact of short-term profit maximization decisions on mid-term scheduling within a compact stochastic mixed integer linear programming (MILP) approach. A test case with a three-stage, 90-scenario stochastic tree and data of the Greek Power System is presented. Results provide insight on both mid-term reservoir management and short-term market-based operation and indicate the feasibility of solving a very large-scale scheduling problem as a unique MILP using a commercial solver. The role of forward contracts in mitigating market power is also analyzed.**

***Index Terms*—Hydro producer, market power, mid-term scheduling, price-maker, pumped storage, stochastic mixed-integer programming.**

NOMENCLATURE

η_k^P	Cycle efficiency of pumped storage plant k.
$\pi_{s,n,t}$	Market clearing price for residual demand curve step n, in \$/MWh.
$\pi_t(q_t)$	Residual demand curve, representing market clearing price as a stepwise monotonically non-increasing function of producer's quota, in \$/MWh.
π_t^{FC}	Forward contract price, in \$/MWh.
$\Pi_{s,t}^{DA}$	Hourly remuneration or payment for the hydro producer in the day-ahead market.
$\mathcal{B}_{s,t}$	Unique scenario bundle for scenario s at hour t.
$D_{s,t}$	Load demand, in MW.
$E[C_T]$	Expected annual profit, in \$.
E_t^{FC}	Forward contract volume, in MWh.
$f_{s,n,t}$	Fraction of the producer's quota for residual demand curve step n, in MWh.
$j(\mathcal{J})$	Index (set) of hydroplants.
$k(\mathcal{K})$	Index (set) of pumped storage plants ($\mathcal{K} \subseteq J$).
$l_j^U(\mathcal{L}_j^U)$	Index (set) of hydroplants/reservoirs immediately upstream of hydroplant/reservoir j.
$n(\mathcal{N}^t)$	Index (set) of residual demand curve steps.
p_s	Probability of scenario s.
$P_{s,j,t}^G$	Power output of hydroplant j, in MW.
$\overline{P_j^G}$	Maximum power output of hydroplant j, in MW.
$P_{s,k,t}^P$	Pumping load of pumped storage plant k, in MW.
$\overline{P_k^P}$	Maximum pumping load of pumped storage plant k, in MW.
$q_{s,t}$	Hydro producer's quota, equal to the producer's total accepted energy offer, for scenario s, in MWh.
$\underline{q}_{s,n,t}$	Minimum producer quota for residual demand curve step n, in MWh.
$R_{s,j,t}$	Hourly net inflow in reservoir j, in hm^3 (MCM).
$s(\mathcal{S})$	Index (set) of scenarios.
sc_j	Specific consumption of hydroplant j, in $\mathrm{hm}^3/\mathrm{MWh}$.
$Sp_{s,j,t}$	Spillage over reservoir j, in hm^3.
$t(\mathcal{T})$	Index (set) of time periods (hours).
$V_{s,j,t}$	Effective volume of water stored in reservoir j at the end of hour t, in hm^3.
$\overline{V_j}$	Effective storage capacity of reservoir j, in hm^3.
$V_{s,j}^{FINAL}$	Target final effective volume for reservoir j, in hm^3.
$V_j^{INITIAL}$	Initial effective volume of reservoir j, in hm^3.
$\boldsymbol{x}_{s,t}$	Vector of the producer's continuous and binary decision variables.
$y_{s,n,t}$	Binary variable indicating the selection of residual demand curve step n.

Manuscript received November 05, 2009; revised June 09, 2010 and October 18, 2010; accepted January 28, 2011. Date of publication March 17, 2011; date of current version October 21, 2011. This work was supported by the State Scholarships Foundation of Greece. Paper no. TPWRS-00864-2009.

The authors are with the Department of Electrical and Computer Engineering, Aristotle University of Thessaloniki, GR-54124 Thessaloniki, Greece (e-mail: cbaslis@ee.auth.gr; bakiana@eng.auth.gr).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TPWRS.2011.2119335

0885-8950/$26.00 © 2011 IEEE
Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 22:18:07 UTC from IEEE Xplore. Restrictions apply.

I. Introduction

OVER the last two decades, power systems worldwide have undergone radical restructuring toward the establishment of efficient and well-functioning competitive electricity markets. While specific legislative changes are still under debate, there is a growing need for new support and decision tools that will help assess the operation and scheduling of independent power producers and power systems as a whole.

In a mid-term perspective, given the fact that a number of market players are competing under profit maximization terms, a wide class of market equilibrium models has been developed [1], [2]. The majority of them are theoretically based on Cournot competition [3], [4], conjectural variation approaches [5], [6], and supply function equilibria (SFE) [7].

An important inherent property of these models is that they provide the interaction among the players' strategies; especially in a mid-term horizon, players are expected to respond to market conditions and properly adjust their policies [2]. However, these market-wide models have certain drawbacks. Cournot-based models deal with competition in quantities; thus, resulting prices depend on demand only and they are frequently higher than those actually observed [1]. In addition, these models depend on demand elasticity, which in turn is difficult to calculate. On the other hand, SFE approaches are computationally hard to deal with, thus necessitating simplifications that result in poor modeling of important operation aspects. A promising alternative to both these methods, especially in the medium term, is the conjectural variation approach, which considers the conjectures made by a producer about the response of his rivals to changes in price [5] or the variation of the price with respect to his own production [6].

Apart from the market-wide equilibrium models, there is a variety of one-firm profit-maximization models that focus on the self-scheduling of a certain power producer, either price-taker [8], or price-maker [9], [10]. In the latter case, the producer's interaction with the rest of the players is usually modeled by residual demand curves [11]. The major advantage here is computational tractability, enabling the detailed and realistic modeling of the physical generation and system operation. Such models are mainly used for short-term scheduling.

The scope of this paper is to present a mid-term self-scheduling method for a price-maker producer that compactly models the impact of the short-term profit maximization decisions on the optimal mid-term operation. Mid-term and short-term operations are usually decoupled, with the latter being driven by specific signals (e.g., water values) derived from the former [12]. As compared to these two-level approaches, the proposed method aims at a much more detailed representation of the short-term operation within the mid-term framework, thus enabling the definition of more realistic mid-term decisions. Our focus is on a hydro producer with significant generating capacity. Such a producer is able to strategically manage his cheap and flexible hydro and pumped-storage plants in the short term, in order to manipulate market prices and maximize profit. Nevertheless, annual water resources are limited. Mid-term water management is thus strongly affected by the short-term profit-based decisions. In addition, various sources of uncertainty are present; among them, stochastic inflows are a key aspect in mid-term planning, due to their seasonal cycling pattern [13], [14]. The proposed model includes inflow uncertainty in terms of a multistage scenario tree.

Short-term price-maker self-scheduling models, which include uncertainty in the residual demand curve, have been proposed in [9] and [10]. In the mid-term, market uncertainty is mainly due to the unknown demand, unit availability, and the response of the producer's competitors. In this paper, uncertainty in the residual demand curve is modeled along with uncertain inflows through the aforementioned scenario tree.

In general, the mid-term self-scheduling problem for a price-maker producer is a large-scale stochastic nonlinear and nonconvex optimization problem. The related technical literature is limited. In a recent effort [15], stochastic dual dynamic programming is applied to solve the problem in the long term, under inflow uncertainty. An approximation of the profit function by its concave hull is though necessary for the implementation of the solution algorithm; this might lead to considerable deviations from the actual shape of the profit function. In this paper, the problem is formulated as a stochastic mixed-integer linear program (MILP) and solved using commercial optimization software (GAMS/CPLEX [16], [17]). The proposed method is tested on a hypothetical hydro producer comprising the major hydroplants of the Greek Power System. The impact of fixed forward contracts is also assessed. Results on reservoir management, market power, market clearing prices, and short-term generation and pumping strategies are presented.

The main contributions of this paper are:

- The optimal mid-term scheduling considering the impact of the short-term profit-based operation under inflow and residual demand uncertainty, for a price-maker hydro producer, within a unique stochastic MILP problem. The hourly time resolution captures the important influence of demand variations and competitors' offers on the producer's bidding strategy, while enabling the detailed modeling of the chronological reservoir operation.
- The introduction of a proper modification of the residual demand curve, which is necessary in order to define the optimal pumped-hydro bidding strategy in the DA market and assess its impact on market prices. Although pumping corresponds to a small fraction of the producer's generation, its influence on price can be significant [18]. Technical literature on optimal pumped-hydro bidding strategies is limited and focuses mainly on short-term operation and Cournot-based [18] or price-taker [19], [20] models. In this paper, the MILP formulation of [21] for the deterministic short-term profit maximization of a price-maker is extended in order to include optimal pumped-hydro bids, through the modified RDC, within a mid-term stochastic programming approach.

II. Model Description and Formulation

The scope of the proposed method is to define the optimal yearly self-scheduling of a price-maker hydro producer under uncertainty. In this section, the basic model characteristics are analyzed.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 22:18:07 UTC from IEEE Xplore. Restrictions apply.

A. Market Modeling—Residual Demand Curve

The hydro producer participates in the day-ahead (DA) energy market, by submitting hourly price-quantity supply offers to the independent system operator (ISO) for each hour of the following day. The DA market is organized as a series of uniform-price single-sided sealed-bid hourly auctions. Single node pricing is assumed. Following the clearing of the DA market by the ISO, the hydro producer is paid for his hourly energy offers that have been accepted at the corresponding system marginal price.

The hydro producer is considered to be a price-maker and, therefore, can influence prices through his bidding strategy. This can be modeled on an hourly basis with the corresponding residual demand curve (RDC), which represents market clearing price (MCP) as a function of the producer's total accepted energy offer, also known as quota. The RDC models MCP as a monotonically non-increasing function of the producer's quota; it is also a stepwise curve, due to the stepwise nature of the supply offers [21]. In general, the RDC provides the interaction of the price-maker producer with all his competitors, the demand, and the energy market legislation.

Since the price-maker hydro producer owns pumped-storage plants, the impact of pumping on market clearing prices must also be modeled. Therefore, a proper modification on the formulation of the RDC is introduced. First, the aggregated bid curve (ABC) is formulated by sorting the competitors' supply bids in ascending order, for the specific hour considered. Fig. 1(a) shows the ABC, as well as its intersection with load demand, which is assumed to be inelastic. Pumping will actually increase the load demand; the augmented load demand is calculated as the sum of the load demand and the producer's total pumping capacity. Fig. 1(b) shows the intersection of the ABC with the augmented load demand. The highlighted part of the ABC between the actual and the augmented load demand in Fig. 1(b) provides market clearing prices for different values of the producer's pumping load (between zero and maximum pumping capacity). Without considering pumping, the RDC would be formulated as the mirror image of the part of the ABC up to the demand D_t with respect to the perpendicular bisector of the interval $[0, D_t]$; the resulting RDC would only comprise the steps denoted with the thin line in Fig. 1(c). The modification we propose is to also include the additional part of the ABC between the actual and the augmented load demand. As denoted with the bold line in Fig. 1(c), this part is placed in the negative quota axis of the RDC. In this way, the effect of pumping on market clearing price is also considered. Pumping is modeled as negative quota, meaning that the producer consumes (instead of producing) energy for water pumping, for which he now pays (instead of being compensated) at the resulting market price.

Forward contracts are also included in the model, in order to evaluate their impact on the self-scheduling of the hydro producer. Forward contracts are not optimized; they are considered as input data with a known price π_t^{FC} (\$/MWh) and volume $E_t^{FC}(MWh)$ for each hour of the yearly horizon. The hydro producer purchases the contracted energy at the market clearing price and sells it at the predefined contract price.

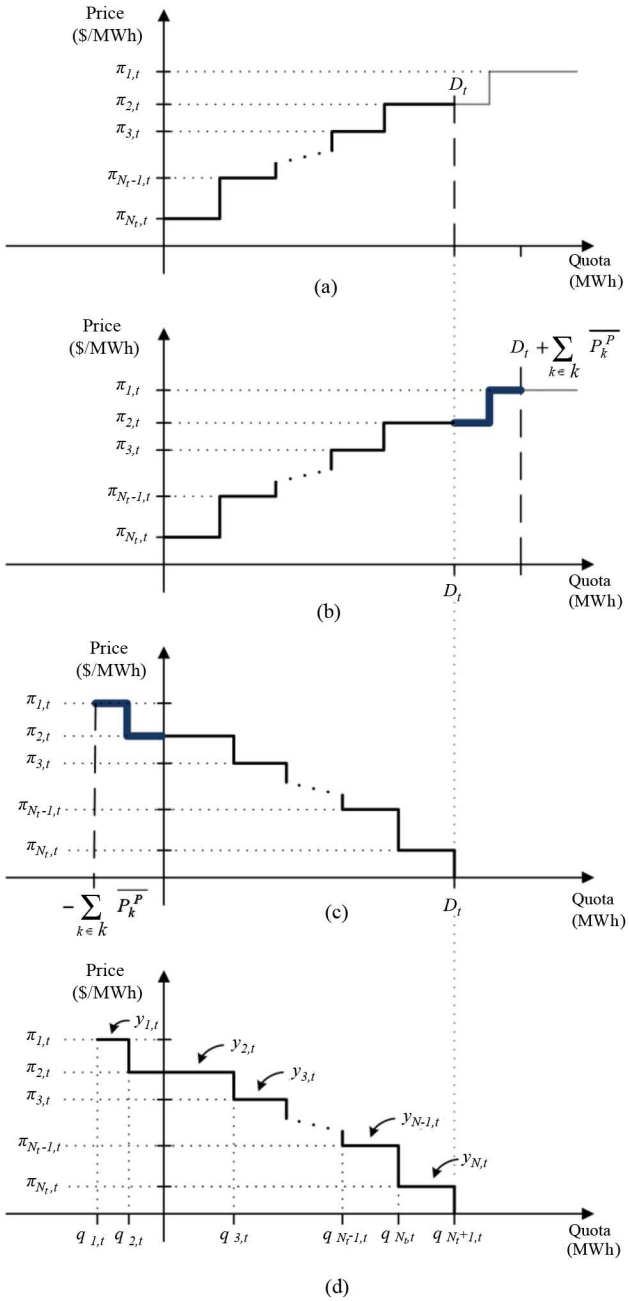


Fig. 1. (a) Aggregated bid curve and load demand. (b) Aggregated bid curve and augmented load demand. (b) Modified residual demand curve. (c) Binary-variable-based linearization [21].

The objective of the hydro producer is to define the optimal hourly quotas that maximize his profit:

$$Profit = \sum_t \pi_t(q_t) \cdot q_t + \left(\pi_t^{FC} - \pi_t(q_t)\right) \cdot E_t^{FC}. \quad (1)$$

The objective (1) is a nonlinear function, due to the product of two variables [quota q_t and the resulting market clearing price $\pi_t(q_t)$]. In order to avoid nonlinearity, the problem is linearized using proper binary variables, one for each step of the RDC, Fig. 1(d). This linearization process has been developed in [21]

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 22:18:07 UTC from IEEE Xplore. Restrictions apply.

and is briefly described in Section II-E. Thus, the deterministic nonlinear profit maximization problem is formulated as a mixed-integer linear program.

B. Hydro Modeling

Reservoir water balance is modeled on an hourly basis, including the explicit consideration of hydraulic coupling among cascaded hydroplants. As already mentioned, pumped storage plants are also modeled. Optimal pumping is calculated on profit maximization criteria only; no predefined rules for the operation of pumped storage plants are assumed.

Hydro unit output power depends on turbine outflow and net head [22]. Many methods of modeling the hydro production function in detail and efficiently including it in relevant scheduling schemes have been presented in the technical literature, for example, [8] and [23]. In this paper, similarly to most mid-term models [14], head dependencies are ignored. Hydro unit output is considered to be a linear function of the turbine outflow rate, assuming a constant specific consumption.

Decisions on the final reservoir volume are generally guided by the results of longer-term scheduling models [14], using either future cost functions or specific target values [24]. In hydro systems comprising reservoirs with yearly utilization cycles, a good practice is to use the corresponding average historical level as a target value for the final reservoir volume. Such an end-of-horizon constraint is included in our model.

C. Stochastic Formulation

Several stochastic parameters affect mid-term operation and scheduling in power systems, including inflows, fuel prices, load demand, unit availability, and producers' market behavior. As operating decisions have to be made in advance, before the realization of the stochastic parameters is observed, an optimal decision strategy that hedges against unknown future must be defined. In this way, some initial decisions are made in view of the yet unknown outcome of the stochastic parameters; as future reveals, the producer's policy is properly adapted.

Stochastic programming models have been extensively used in a wide range of real-life problems in order to calculate optimal decision strategies that hedge against uncertainty. Uncertainty is usually considered to be fully described by a finite number of possible realizations of the unknown parameters along the planning horizon (with their corresponding probabilities) called scenarios. The scenarios are combined to form the scenario tree. Stages are used to define when a new realization of the stochastic parameter is observed and the related information is available for decision making. Fig. 2 illustrates a three-stage scenario tree, with six scenarios in total. The summation of the unconditional probabilities of the six scenarios is equal to 1. Optimal decisions are based on the information available at the time of decision and should not anticipate any certain realization of the stochastic parameters in the future. Therefore, if two or more scenarios are identical up to time period t (Fig. 2), all decisions regarding these scenarios must also be identical, for all time periods up to t. This requirement is known as the principle of non-anticipativity [25] and is of vital importance in stochastic programming. Non-anticipative

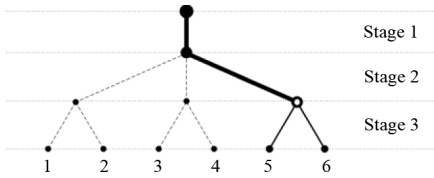


Fig. 2. Three-stage scenario tree. Scenarios 5 and 6 are identical for the first and the second stage.

decisions are the basis of a well-hedged scheduling strategy; when uncertainty is revealed, the producer will be able to efficiently respond even to the worst scenario.

In the proposed model, stochastic reservoir inflows are considered. Inflows are a key aspect in mid-term operation; decisions on reservoir management are coupled in time and thus highly dependent on the unknown future inflows. Under a strategy that is not well-hedged against inflow uncertainty, reservoirs are exposed to the danger of undesirable emptying or spillage, in the event of a highly dry or wet period, respectively [24]. Reservoir inflows (mainly originated from rainfall and snow melting) have seasonal cycling characteristics and therefore they largely affect mid-term scheduling. In the short term, a price-maker hydro producer maximizes his profit under the corresponding bidding strategy for the DA market. When load demand is high, the producer may decide to either release water and profit from the corresponding high marginal prices or to withhold capacity in order to provoke price spikes. Since reservoir inflows are the unique source of energy for the hydro producer, a mid-term scheduling model that considers the stochastic nature of inflows is very important for the aforementioned water usage strategy in terms of profit maximization. Such a mid-term model can also capture the seasonal characteristics of load demand (e.g., summer peaks), that greatly influence the producer's short-term strategy.

The rest of the unknown parameters, namely unknown fuel prices, load demand, as well as competitors' unit availability and bidding strategies, can be compactly modeled by a stochastic residual demand curve [9], [10]. In this paper, a scenario is defined by the combined uncertainties of inflows and residual demand curves.

A three-stage scenario tree with inflow and demand uncertainty is used, Fig. 3. The proper selection of the number and duration of the stages strongly depends on the properties of the underlying uncertain parameters as well as the starting time point and the duration of the planning horizon [26]. The three-stage scenario tree of Fig. 3 adapts to a power system with two high inflow and two high demand periods; each of the second and the third stage include a main inflow and a main demand period.

A deterministic first stage of one month is adopted, as forecasts are considered to be accurate within this period. The second stage comprises the next four months. Five inflow realizations are modeled, namely high (H), medium-high (MH), medium (M), medium-low (ML), and low (L). Each inflow realization is combined with three demand realizations considered, namely high (h), medium (m), and low (l), to formulate the 15 second-stage branches of the scenario tree. A specific

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 22:18:07 UTC from IEEE Xplore. Restrictions apply.

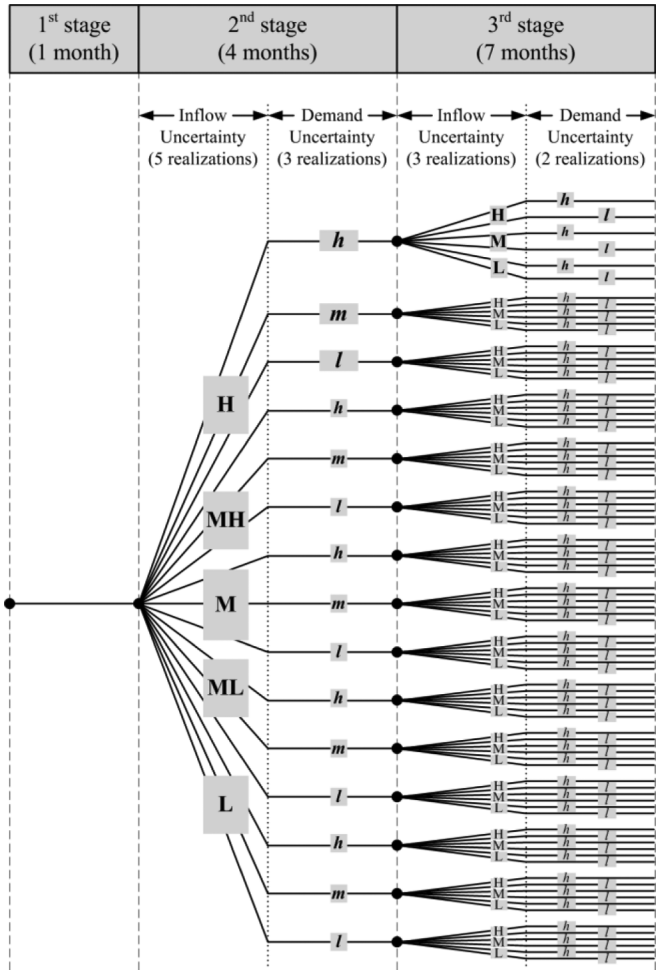


Fig. 3. Yearly three-stage 90-scenario tree.

inflow/demand realization is modeled through a scale factor which modifies the expected hourly inflow/demand curve of the second stage. In order to formulate the stochastic RDC, we express the correlation between the observed inflow and demand realization and the conjectured reaction of the producer's competitors to it. For this purpose, the competitors' expected bid prices are properly modified by using a corresponding scale factor. In general, higher/lower bid prices are expected during dry/wet periods. On the other hand, competitors are expected to submit higher/lower bid prices with higher/lower demand realizations (for example, under a lower demand scenario, they are likely to act more competitively and thus lower their bid prices, in order to maintain their market share). Stochastic RDC realizations are thus formulated for the second stage, based on the respective demand realization and the conjectured modification of the competitors' bid prices, following the process of Section II-A.

Finally, the duration of the third stage is longer, equal to the remaining seven months. For each one of the 15 second-stage parent nodes there are six branches, as defined by the combination of three possible inflow realizations (high, medium, low) and two demand realizations (high, low), Fig. 3. These third-stage realizations can be calculated with respect to their parent realization in the second stage, thus modeling the possible correlation between second- and third-stage stochastic parameters. Therefore, the designation of a third-stage branch in Fig. 3 is relevant to its parent second-stage branch; third-stage branches with the same designation (e.g., "H" for high inflows) do not necessarily refer to realizations of the same magnitude. The formulation of third-stage stochastic RDCs follows the procedure already explained for the second stage.

The proposed model can accommodate a more representative scenario tree, derived from more accurate seasonal forecasting techniques [26]. Estimating the residual demand curves is also a cumbersome task that requires processing of historical demand and competitors' behavior data [9]. Scenario reduction, based on probability metrics, is another important tool for constructing a tractable, yet representative, smaller-sized scenario tree [27].

D. Scope of the Stochastic Mid-Term Scheduling Model

The proposed model is developed as a mid-term scheduling tool and does not aim at defining the optimal short-term operation of the producer. If one seeks to solve the stochastic short-term problem, he must take into account hourly RDC scenarios for the next day and define the optimal multi-step, nondecreasing bid curves that hedge against day-ahead uncertainty, as in [9]. In our approach, RDC uncertainty is considered in the mid-term and refers to the increasing/decreasing trend of demand within each stage and the expected mid-term adjustment of the competitors' strategies to the observed inflow and demand conditions. For a specific scenario, stage, and hour, one corresponding hourly RDC is used in order to describe the respective hourly market conditions; that is, within each stage, the producer holds a deterministic view over the market parameters of this stage. Thus, as analyzed in [28], the optimal bidding strategy is defined by one price-quantity pair for each hour. Hence, the producer's bidding strategy over the mid-term horizon consists of one such price-quantity pair for each hour of each scenario. In fact, what is defined through the RDC is the optimal quantity offer; the optimal bid price can then be any price which is lower than or equal to the market price derived from the RDC [29].

The hourly time resolution is adopted because profit is made on an hourly basis, where a slight change in the producer's offered quantity may have a big impact on hourly market price and profit. In addition, it enables modeling the hourly chronological reservoir operation. Alternatively, the aggregation of time periods with similar market conditions can be chosen, in order to reduce the problem size and increase the stochastic modeling detail. An example of monthly aggregate RDCs in a mid-term price-maker problem can be found in [15].

The adopted hourly detail also allows a deeper insight on how pumping influences market prices and increases the producer's profit. Pumping cycles are usually daily or weekly; since they are price-sensitive though, they also depend on the specific seasonal demand and inflow conditions. In addition, when a pumped-storage plant is coupled with a large upstream reservoir, the seasonal regulation of the latter may largely affect pumping (e.g., as with excessive upstream water flow).

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 22:18:07 UTC from IEEE Xplore. Restrictions apply.

E. Stochastic Profit Maximization Problem Formulation

The objective of the producer is to maximize the expected annual profit. The stochastic mixed integer linear problem is formulated as follows:

$$\text{Maximize } E[C_T] = \sum_{s\in\mathcal{S}} p_s \cdot \sum_{t\in T} \Pi^{DA}_{s,t} + \pi^{FC}_t \cdot E^{FC}_t \quad (2)$$

$$\text{Subject to } \Pi^{DA}_{s,t} = \sum_{n\in\mathcal{N}^t} \pi_{s,n,t} \cdot \left(f_{s,n,t} + \underline{q}_{s,n,t} \cdot y_{s,n,t}\right) - \sum_{n\in\mathcal{N}^t} \pi_{s,n,t} \cdot y_{s,n,t} \cdot E^{FC}_t, \forall s,t \quad (3)$$

$$q_{s,t} = \sum_{j\in\mathcal{J}} P^G_{s,j,t} - \sum_{k\in\mathcal{K}} P^P_{s,k,t}, \ \forall s,t \quad (4)$$

$$q_{s,t} = \sum_{n\in\mathcal{N}^t} f_{s,n,t} + \underline{q}_{s,n,t} \cdot y_{s,n,t}, \forall s,t \quad (5)$$

$$\sum_{n\in\mathcal{N}^t} y_{s,n,t} = 1, \ \forall s,t \quad (6)$$

$$0 \le f_{s,n,t} \le (\underline{q}_{s,n+1,t} - \underline{q}_{s,n,t}) \cdot y_{s,n,t}, \ \forall s,n,t \quad (7)$$

$$V_{s,j,t} = V_{s,j,t-1} - sc_j \cdot P^G_{s,j,t} - Sp_{s,j,t} + \eta^P_j \cdot sc_j \cdot P^P_{s,j,t} + R_{s,j,t} + \sum_{l^U_j\in\mathcal{L}^U_j} sc_{l^U_j} \cdot P^G_{s,l^U_j,t} + Sp_{s,l^U_j,t} - \eta^P_{l^U_j} \cdot sc_{l^U_j} \cdot P^P_{s,l^U_j,t}, \forall s,j,t \quad (8)$$

$$0 \le P^G_{s,j,t} \le \overline{P^G_j}, \ \forall s,j,t \quad (9)$$

$$0 \le P^P_{s,k,t} \le \overline{P^P_k}, \ \forall s,k,t \quad (10)$$

$$0 \le V_{s,j,t} \le \overline{V_j}, \ \forall s,j,t \quad (11)$$

$$V_{s,j,0} = V^{INITIAL}_j, \ \forall s,j \quad (12)$$

$$V_{s,j,T} = V^{FINAL}_{s,j}, \ \forall s,j \quad (13)$$

$$\boldsymbol{x}_{s_1,t} = \boldsymbol{x}_{s_2,t}, \ \forall s_1,s_2,t : \mathcal{B}_{s_1,t} = \mathcal{B}_{s_2,t}. \quad (14)$$

Hydro producer profit is expressed in (2)–(7). For each scenario, profit is multiplied by the scenario probability. The producer is compensated for his produced energy at the market clearing price. When pumping, the producer buys at the market clearing price. On the other hand, the producer buys his contracted energy from the ISO at the market clearing price and, in turn, sells it at the respective forward contract price.

Profit is derived from the modified residual demand curve, Fig. 1(d). Producer's quota is expressed in (4); pumping is modeled as negative quota. Constraints (5)–(7) correspond to the binary-variable-based linearization of the profit maximization problem [21]. In brief, for a given hour t and scenario s, only one binary variable $y_{s,n,t}$ can be activated, thus indicating the selection of the corresponding RDC step n, (6). Given the selected step, constraint (5) determines the exact value of the producer's quota $q_{s,t}$, as follows. The minimum quota for step n is defined by the parameter $\underline{q}_{s,n,t}$, whereas the nonnegative continuous variable $f_{s,n,t}$ represents the additional fraction of the quota within the step n. The upper bound of $f_{s,n,t}$ is the difference between the minimum quota parameters of the next $(n+1)$ and the current (n) step, (7); if step n is not selected $(y_{s,n,t} = 0)$, then the upper bound of $f_{s,n,t}$ is zero and thus $f_{s,n,t} = 0$.

TABLE I
HYDRO SYSTEM DATA

River Basin	Number of Hydroplants (Pumped-Storage Plants)	Capacity (Pumping), MW
Aliakmon	3 (1)	800 (330)
Aheloos	2	757
Arachthos	2	544
Nestos	2 (1)	491 (390)
Tavropos	1	130

Market clearing prices for each step of the curve, $\pi_{s,n,t}$, are known in advance and therefore no product of unknown variables exists in the profit function (3). Price is implicitly defined by the activation of the corresponding binary variable $y_{s,n,t}$. The required payment for the purchase of the forward contract energy from the ISO is modeled in the same fashion; the contracted energy E^{FC}_t, which is a known parameter, is multiplied with the implicitly defined market price; see (3). The producer's forward contract revenue is a constant term and is included in (2).

Following the modification of the RDC proposed in this paper, a minor adjustment of the linearization process of [21] is necessary. In particular, the minimum quota parameters $\underline{q}_{s,n,t}$ which correspond to pumping, i.e., to the left-most, negative quota segments of the modified RDC, Fig. 1(d), must be negative in order to enable negative (pumping) quotas through (5) and (7). On the contrary, the respective minimum quota parameters in [21] are strictly nonnegative.

Hourly reservoir water balance is expressed in (8), ignoring water travel time between reservoirs. The rest of the hydro producer's operating constraints include hydroplant maximum power output (9), maximum pumping limit (10), reservoir volume upper bound (11), initial value (12), and final value (13).

Non-anticipativity is expressed in (14). A scenario bundle is defined as a subset of the scenario space $\mathcal{S}$ containing scenarios that are identical up to a certain time [25]. For example, in Fig. 2, a bundle is the subset comprising scenarios 5 and 6, as the latter share the same information up to the end of stage 2 (bold solid line). At a certain time t, each scenario s belongs to exactly one bundle, denoted as $\mathcal{B}_{s,t}$. According to (14), if two scenarios belong to the same bundle at time t, i.e., are identical up to t, the corresponding operating decisions should be identical, too.

III. TEST RESULTS

The proposed method was applied to a hydro system configuration comprising all major hydroplants and reservoirs of the Greek Power System. As presented in Table I, the hypothetical hydro producer owns ten hydroplants, including two pumped storage plants, in five river basins and four major blocks of reservoirs in cascade. The producer's total generating capacity is 2720 MW (total system capacity is about 9500 MW), whereas pumping capacity is 720 MW. The cycle efficiency of pumped storage plants is assumed to be 75%.

The yearly planning horizon starts in October; the first stage comprises October, the second stage includes the next four months, from November to February and the third stage refers to the last seven months, from March to September. This stage configuration was based on the hydrologic and load demand pattern of Greece. High inflows are observed in winter and

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 22:18:07 UTC from IEEE Xplore. Restrictions apply.

TABLE II
MEAN INFLOW CHARACTERISTICS

Total inflows (GWh)	Winter	Spring	Summer	Autumn
3998	35.3%	45.3%	7.4%	12.0%

TABLE III
SYSTEM LOAD CHARACTERISTICS

Total annual demand (GWh)	Peak load (MW)	Base load (MW)	Load factor
47260	8888	2764	0.607

TABLE IV
INFLOW REALIZATIONS

Stage	2					3		
Realization	H	MH	M	ML	L	H	M	L
Factor	1.5	1.25	1	0.75	0.5	1.25	1	0.75
Probability	0.1	0.2	0.4	0.2	0.1	0.25	0.5	0.25

TABLE V
DEMAND REALIZATIONS

Stage	2			3	
Realization	h	m	l	h	l
Factor	1.1	1	0.9	1.1	0.9
Probability	0.25	0.5	0.25	0.5	0.5

TABLE VI
COMPETITORS' BID PRICE MODIFICATION

Stage	2														
Inflow	H			MH			M			ML			L		
Demand	h	m	l	h	m	l	h	m	l	h	m	l	h	m	l
Factor	1	0.95	0.925	1	0.975	0.95	1.05	1	0.975	1.15	1.05	1	1.25	1.15	1
Stage	3														
Inflow	H						M						L		
Demand	h			l			h			l			h		l
Factor	1			0.95			1.05			0.975			1.15		1

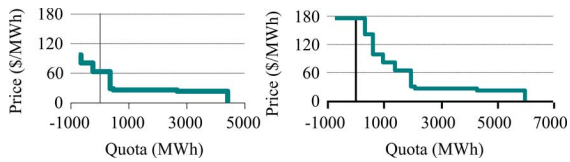


Fig. 4. Typical residual demand curves for low and high demand hours (left and right shape, respectively).

TABLE VII
FORWARD CONTRACT TEST CASES

Test Case	Hourly Volume (MW)		Hourly Price ($/MWh)	
	10pm – 8am	9am – 9pm	10pm – 8am	9am – 9pm
A	125	250	50	100
B	250	500		
C	500	1000		

spring months; the former are included in the second stage and the latter in the third stage (summer and autumn are much drier). On the other hand, peak load is observed during summer and is included in the third stage, while winter demand is higher than autumn and spring and is included in the second stage. Average inflow information is shown in Table II; yearly inflows are 4 TWh, twice the total storage capacity of the reservoirs. A chronological average inflow curve was derived from the typical hydrology of Greece. Final reservoir target volume $V_{s,j}^{FINAL}$ was set equal to the initial value $V_j^{INITIAL}$, for all scenarios s. A typical yearly chronological demand curve was formulated from data published by the Hellenic TSO [30]. A basic overview is shown in Table III. Note that system reserves are low; peak load is 8.9 GW while total capacity is 9.5 GW.

Both the inflow and the demand curves served as the basis for the formulation of the 90-scenario stochastic tree (Fig. 3), following the procedure presented in Section II-C. For each possible inflow/demand realization, a corresponding scale factor was used to modify the average inflow/demand curve, as presented in Tables IV and V. In order to express the high dependence of (third-stage) spring inflows on (second-stage) winter inflows within a specific scenario, the scale factor of the third-stage inflow realization shown in Table IV was multiplied with the scale factor of its parent second-stage realization. This product was then used as the actual scale factor for the third-stage inflows of the specific scenario. The same approach was adopted for the third-stage demand realizations.

The conjectured response of the producer's competitors to each inflow and demand realization, which was discussed in Section II-C, is modeled using the scale factors of Table VI. These factors were used in order to modify the estimated bid price offers of the competitors during the corresponding stage. The estimation of the competitors' offers was made by taking into account realistic data from the Greek Power System (all competitors are thermal producers and their bids were assumed to reflect the marginal cost of their units).

For each scenario and each hour of the planning horizon, one separate RDC was calculated from the corresponding scenario-dependent competitors' offers and hourly demand. Two such curves are presented in Fig. 4 for low and high demand hours. When demand exceeds the competitors' total installed capacity, the first RDC step is equal to the competitors' supply deficit and is priced at the price cap (175 $/MWh). RDCs of up to eight price levels were used in our tests. This number was considered sufficient for representing the main generating unit technologies of the Greek Power System. In energy markets with a higher degree of diversification of generating technologies, large number of pricing zones or intense strategic behavior among the participants, a larger number of price levels may be required. Time block aggregation or reduction of the scenario tree would probably be necessary in such a case, since there is a trade-off regarding the detailed modeling of chronological market operation, the uncertainty, and the RDC.

A scenario is defined by the combination of inflow and residual demand uncertainties; thus, the unconditional scenario probability is calculated as the product of the conditional probabilities of the respective realizations, presented in Tables IV and V. Finally, three test cases of fixed forward contracts were examined, as in Table VII. The producer delivers the specified contract energy at the corresponding contract price.

The model was implemented in GAMS/CPLEX 12 on an Intel Core i7 2.8-GHz PC with 24 GB of RAM under 64-bit Windows 7. Table VIII presents information about the number of constraints and variables in our model, as well as the execution time for the no-contract test case. Similar run times,

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 22:18:07 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
MODEL CHARACTERISTICS

Equations	9,583,847
Variables	23,217,165
Integer variables	3,509,638
Total run time	~ 28 hours

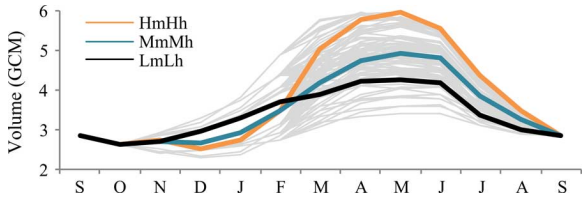


Fig. 5. Total reservoir storage at the end of each month for all scenarios.

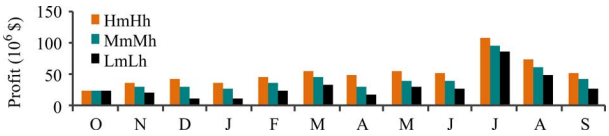


Fig. 6. Monthly producer profits for scenarios HmHh, MmMh, and LmLh.

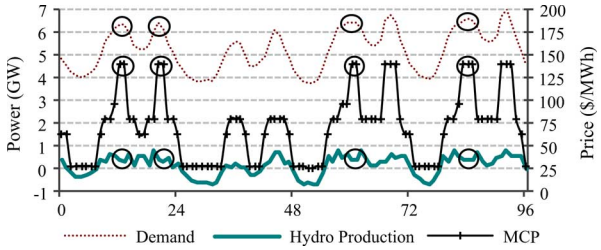


Fig. 7. Short-term overview of hourly load demand, market clearing price, hydro production, pumping, and capacity withholding.

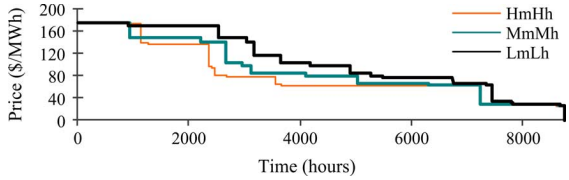


Fig. 8. Annual price duration curves for scenarios HmHh, MmMh, and LmLh.

around 30 h, were required for the forward contract test cases as well. Parallel processing within the CPLEX solver (by simultaneously using more than one PC thread) can be used to accelerate the solution. However, this may increase memory requirements. Obviously, the proposed stochastic model results in a very large-scale MILP problem with intense computing requirements. This is why large physical memory and a 64-bit operating system are of paramount importance (the memory dedicated to GAMS/CPLEX during the solution was almost never below 15 GB, with a maximum of 20.2 GB). Despite the size of the problem (23 million variables, out of which 3.5 million are integer), the solution obtained for all test cases was within an optimality gap of 0.3% (for the no-contract test case the gap was as small as 0.01%). Therefore, the solution of the yearly stochastic self-scheduling of a midsized price-maker hydro producer as a unique MILP problem without the use of decomposition techniques [10] has proved to be a feasible choice.

Optimal reservoir management results are presented in Fig. 5, where three scenarios, namely HmHh, MmMh, and LmLh, have been highlighted. Uppercase letters symbolize the inflow realizations while lowercase letters are used for the demand realizations; second-stage realizations are marked first. Thus, the above three scenarios correspond to the highest (wet), average, and lowest (dry) inflows under average and high demand in the second and the third stage, respectively.

As shown in Fig. 5, the general pattern of mid-term reservoir management is to gradually increase water storage until the beginning of summer (quite similarly to a centralized, cost-minimization approach [31]). Maximum storage for almost all scenarios is observed in May. The main seasonal motivation for the hydro producer is to use the accumulated water in order to profit from the high prices due to the summer peak load (mainly observed in July and August). Monthly profits are presented in Fig. 6; profits in July and August are higher than in the rest of the year.

The optimal mid-term decisions of the producer are defined within the framework of profit maximization. Since profit is generated at the hourly market operation level, it is clear that the optimal mid-term scheduling depends on the market behavior of the producer in the short term. A further insight is provided in Fig. 7, which presents the producer's self-scheduling within a four-day first-stage (October) interval. Pumping is plotted as negative production; it is scheduled during low demand hours with prices around 25 $/MWh. Although peak load is moderate (below 7 GW), price spikes of 140 $/MWh are observed as the producer withholds capacity and benefits from the resulting increased prices. In Fig. 7, circles are used to indicate the specific peak-load hours during which the producer reduces his production in order to provoke price spikes, defined here as prices higher than 140 $/MWh. Thus, the mid-term scheduling of the producer is also guided by a capacity withholding strategy performed in the short term.

Capacity withholding is performed independently of the inflow realization, as Fig. 8 and Table IX show. The frequency of price spikes is very high; even for the most favorable inflow scenario, they are observed for about 2400 h—more than three months. In addition, although higher inflows result in higher total production, the average production at price spikes remains below 1 GW, while the installed capacity is 2.7 GW. This means that despite the increased inflows, the producer continues to withhold capacity during peak hours and consumes the additional hydro resources during off-peak hours. This increased allocation of hydro generation to off-peak periods is a mid-term manifestation of market power and has been widely observed and analyzed in the technical literature [32]. For a price-taker hydro producer, an opposite management of water resources would be optimal: he would be motivated to allocate more hydro generation to peak periods, thus profiting from the corresponding high prices.

Our analysis has so far focused on the effect of different inflow profiles; relevant comments can be made for the impact of uncertain residual demand as well. Fig. 9 presents monthly profits and optimal production and pumping for two scenarios

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 22:18:07 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
PRICE SPIKE DURATION AND CORRESPONDING PROFIT AND PRODUCTION

Scenario	Price Spike Duration (hours)	Average Production at Price Spikes (MW)	Total Production (GWh)
HmHh	2372	890.2	6492
MmMh	2660	851.1	4049
LmLh	3173	742.7	2711

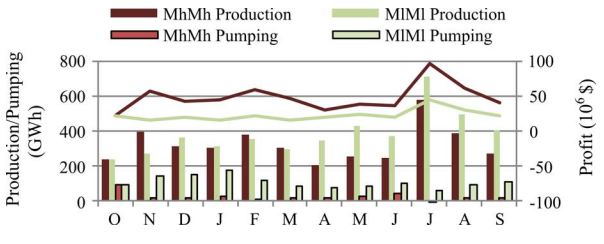


Fig. 9. Production, pumping, and profits for scenarios MhMh and MlMl.

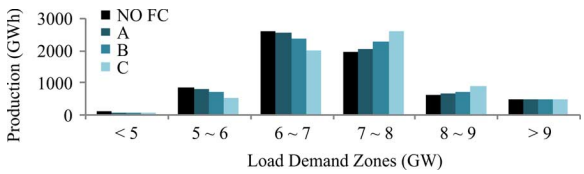


Fig. 10. Load demand levels and hydro production distribution for the forward contract test cases (scenario HmHh).

of medium inflows but opposite demand realizations, namely MhMh and MlMl. Under low demand pumping is utilized to a much higher extent; especially during the second stage, a large part of the production is due to cycling the pumped water instead of releasing water from inflows. Storing the inflow resources increases the reservoirs' content. As a result, more water is transferred to the third stage, thus leading to high levels of third-stage hydro production. On the other hand, under high demand, the producer achieves larger profits with lower production levels due to the higher prices provoked.

To assess the impact of forward contracts on the mid-term scheduling of the producer, three test cases of contracts were examined, as in Table VII. The optimal distribution of hydro production along specific load levels is shown in Fig. 10. Production in the peak-load levels increases with the contract volume; in other words, more hydro resources are allocated from off-peak to peak-load periods. This is a clear indication of market power mitigation. Forward contracts motivate the price-maker producer to lower market clearing prices, at which prices he must now buy the contracted energy from the ISO. As contract volume increases, the incentive for the producer to lower the price and behave more competitively becomes stronger. Under test case C, where the producer is overcontracted, price spikes are limited even for a very dry year with high summer demand (scenario LmLh), Fig. 11.

Fig. 12 presents the optimal water management for the contract test case B; there is a larger deviation among the individual scenario curves than in the no-contract case (Fig. 5). This is due to the price reduction incentive, imposed by the contract, which leads to two alternative ways of managing second-stage stored volume. For low inflow scenarios and/or scenarios with

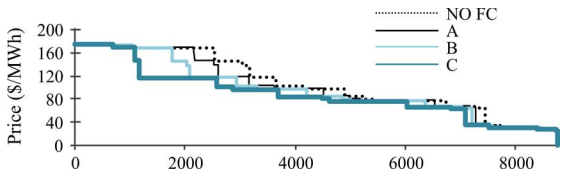


Fig. 11. Price duration curves for the forward contract test cases (scenario LmLh, lowest inflows, and high demand during summer).

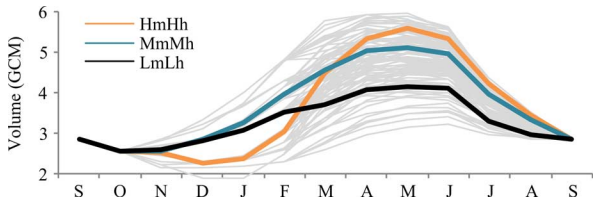


Fig. 12. Total reservoir storage for forward contract test case B.

TABLE X
PUMPING STRATEGY UNDER HIGH VOLUME FORWARD CONTRACTS

Strategy	Pumping		No Pumping	
Hour	3 am	4 am	3 am	4 am
Production (MWh)	-63	47	0	0
MCP ($/MWh)	64.59	29.81	64.59	64.59
FC Volume (MWh)	500	500	500	500
FC Price ($/MWh)	50	50	50	50
Spot Market Revenue ($)	-36365	-13504	-32296	-32296
Contract Revenue ($)	25000	25000	25000	25000
Net Revenue ($)	-11365	11496	-7296	-7296
Total Revenue ($)	131		-14592	

high summer demand, more water is stored to be used during summer peaks; stored volume at the end of the second stage increases. On the contrary, when scenario inflows are high enough to mitigate summer price spikes and/or winter demand is high, more water is consumed in the second stage in order to lower the respective prices; stored volume at the end of the second-stage volume is reduced.

Finally, an interesting example of how pumping strategy can be influenced by forward contracts of high volume is shown in Table X. The column designated "Pumping" corresponds to the optimal strategy obtained from test case C. For the two-night-hour interval, the producer pumps at the high price of 64.59 $/MWh during the first hour and then uses the pumped energy (reduced by the cycle efficiency factor) for hydro production, thus lowering the price of the second hour. In this way, although paying for both the contracted and the pumped energy at the high price of the first hour, he benefits from purchasing the contracted energy at a significantly reduced price during the second hour. In addition, due to a full pumping cycle being performed, no actual inflow resource is consumed. If an alternative "No Pumping" strategy were followed, then the producer would have to purchase the contracted energy of the second hour at the high price, too; this would result in loss, as Table X shows. Therefore, although it might appear as a paradox, under contracts of high volume, it can be optimal to pump at high price and produce at low price. However, this rather extreme behavior is observed to a limited extent, mainly under low inflow scenarios.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 22:18:07 UTC from IEEE Xplore. Restrictions apply.

IV. Conclusion

A yearly self-scheduling model for a price-maker hydro producer that hedges against uncertain reservoir inflows and residual demand curves has been presented. The model was formulated and solved as a large-scale stochastic mixed-integer program. A test case of a hypothetical hydro producer comprising the major hydroplants of the Greek Power System was solved, considering a three-stage scenario tree with 90 scenarios. Results in the mid-term time scale present the optimal reservoir management; in the short term, they illustrate the use of market power through capacity withholding. The impact of fixed forward contracts on market power mitigation was also assessed. The optimization of mid-term decisions considering the short-term bidding strategy within a compact stochastic MILP approach, as well as the proposed modification of the residual demand curve in order to define the producer's optimal pumped-hydro bids, are the main contributions of this paper. The solution of a unique very large-scale MILP problem with a sophisticated commercial solver has proved to be a feasible choice. Advances in computer hardware may allow the solution of even larger problems, including more scenarios and RDC details, without the need to resort to decomposition approaches.

References

[1] E. P. Kahn, "Numerical techniques for analyzing market power in electricity," *Elect. J.*, vol. 11, pp. 34–43, Jul. 1998.

[2] M. Ventosa, A. Baíllo, A. Ramos, and M. Rivier, "Electricity market modeling trends," *Energy Policy*, vol. 33, no. 7, pp. 897–913, May 2005.

[3] T. J. Scott and E. G. Read, "Modelling hydro reservoir operation in a deregulated electricity market," *Int. Trans. Oper. Res.*, vol. 3, no. 3–4, pp. 243–253, Aug. 1996.

[4] L. A. N. Barroso, M. H. C. Fampa, R. Kelman, M. V. F. Pereira, and P. Lino, "Market power issues in bid-based hydrothermal dispatch," *Ann. Oper. Res.*, vol. 117, pp. 247–270, Nov. 2002.

[5] C. J. Day, B. F. Hobbs, and J.-S. Pang, "Oligopolistic competition in power networks: A conjectured supply function approach," *IEEE Trans. Power Syst.*, vol. 17, no. 2, pp. 597–607, May 2002.

[6] E. Centeno, J. Reneses, and J. Barquín, "Strategic analysis of electricity markets under uncertainty: A conjectured-price-response approach," *IEEE Trans. Power Syst.*, vol. 22, no. 1, pp. 423–432, Feb. 2007.

[7] R. J. Green and D. M. Newbery, "Competition in the British electricity spot market," *J. Pol. Econ.*, vol. 100, pp. 929–953, Oct. 1992.

[8] A. J. Conejo, J. M. Arroyo, J. Contreras, and F. A. Villamor, "Self-scheduling of a hydro producer in a pool-based electricity market," *IEEE Trans. Power Syst.*, vol. 17, no. 4, pp. 1265–1271, Nov. 2002.

[9] Á. Baíllo, M. Ventosa, M. Rivier, and A. Ramos, "Optimal offering strategies for generation companies operating in electricity spot markets," *IEEE Trans. Power Syst.*, vol. 19, no. 2, pp. 745–753, May 2004.

[10] M. P. Nowak, R. Schultz, and M. Westphalen, "A stochastic integer programming model for incorporating day-ahead trading of electricity into hydro-thermal unit commitment," *Optimiz. Eng.*, vol. 6, pp. 163–176, Jun. 2005.

[11] G. B. Sheblé, *Computational Auction Mechanisms for Restructured Power Industry Operation*. Norwell, MA: Kluwer, 1999.

[12] J. Reneses, E. Centeno, and J. Barquín, "Coordination between medium-term generation planning and short-term operation in electricity markets," *IEEE Trans. Power Syst.*, vol. 21, no. 1, pp. 43–52, Feb. 2006.

[13] A. Ramos, S. Cerisola, Á. Baíllo, and J. M. Latorre, "Stochasticity in electric energy systems planning," *Lecture Notes Econ. Math. Syst.*, vol. 581, pp. 217–239, 2006.

[14] S. W. Wallace and S. E. Fleten, "Stochastic programming models in energy," in *Stochastic Programming, Handbooks in Operations Research and Management Science*, A. Rusczynski and A. Shapiro, Eds. New York: Elsevier, 2003, vol. 10, pp. 637–677.

[15] B. C. Flach, L. A. Barroso, and M. V. F. Pereira, "Long-term optimal allocation of hydro generation for a price-maker company in a competitive market: Latest developments and a stochastic dual dynamic programming approach," *IET Gen. Transm. Distrib.*, vol. 4, pp. 299–314, Feb. 2010.

[16] A. Brooke, D. Kendrick, A. Meeraus, and R. Raman, *GAMS: A User's Guide*. Washington, DC: GAMS Development, 1998.

[17] ILOG CPLEX 11.0 User's Manual. Gentilly, France, ILOG SA, 2007.

[18] C. Crampes and M. Moreaux, "Pumping water to compete in electricity markets," in *Working Papers 08.13.257, LERNA*. Toulouse, France: University of Toulouse, 2008.

[19] E. Ni and P. B. Luh, "Optimal integrated bidding and hydrothermal scheduling with risk management and self-scheduling requirements," *IEEE Trans. Power Syst.*, vol. 19, no. 1, pp. 600–609, Feb. 2004.

[20] N. Lu, J. H. Chow, and A. A. Desrochers, "Pumped-storage hydroturbine bidding strategies in a competitive electricity market," *IEEE Trans. Power Syst.*, vol. 19, no. 2, pp. 834–841, May 2004.

[21] S. de la Torre, J. M. Arroyo, A. J. Conejo, and C. Contreras, "Price maker self-scheduling in a pool-based electricity market a mixed-integer LP approach," *IEEE Trans. Power Syst.*, vol. 17, no. 4, pp. 1037–1042, Nov. 2002.

[22] A. J. Wood and B. F. Wollenberg, *Power Generation, Operation and Control*. New York: Wiley, 1996.

[23] A. L. Diniz and M. E. P. Maceira, "A four-dimensional model of hydro generation for the short-term hydrothermal dispatch problem considering head and spillage effects," *IEEE Trans. Power Syst.*, vol. 23, no. 3, pp. 1298–1308, Aug. 2008.

[24] M. V. F. Pereira, "Optimal stochastic operations scheduling of large hydroelectric systems," *Int. J. Elect. Power Energy Syst.*, vol. 11, pp. 161–169, 1989.

[25] R. T. Rockafellar and R. J. Wets, "Scenarios and policy aggregation in optimization under uncertainty," *Math. Oper. Res.*, vol. 16, pp. 119–147, Feb. 1991.

[26] J. Jacobs, G. Freeman, J. Grygier, D. Morton, G. Schultz, K. Staschus, and J. Stedinger, "SOCRATES: A system for scheduling hydroelectric generation under uncertainty," *Ann. Oper. Res.*, vol. 59, pp. 99–133, 1995.

[27] N. Gröwe-Kuska, H. Heitsch, and W. Römisch, "Scenario reduction and scenario tree construction for power management problems," in *Proc. IEEE Power Tech Conf.*, Jun. 2003, vol. 3, pp. 23–26.

[28] A. G. Bakirtzis, N. P. Ziogos, A. C. Tellidou, and G. A. Bakirtzis, "Electricity producer offering strategies in day-ahead energy market with step-wise offers," *IEEE Trans. Power Syst.*, vol. 22, no. 4, pp. 1804–1818, Nov. 2007.

[29] C. K. Simoglou, P. N. Biskas, and A. G. Bakirtzis, "Optimal self-scheduling of a thermal producer in short-term electricity markets by MILP," *IEEE Trans. Power Syst.*, to be published.

[30] Hellenic Transmission System Operator S.A. [Online]. Available: http://www.desmie.gr.

[31] C. G. Baslis, S. E. Papadakis, and A. G. Bakirtzis, "Simulation of optimal medium-term hydro-thermal system operation by grid computing," *IEEE Trans. Power Syst.*, vol. 24, no. 3, pp. 1208–1217, Aug. 2009.

[32] J. Bushnell, "A mixed complementarity model of hydrothermal electricity competition in the western united states," *Oper. Res.*, vol. 51, pp. 80–93+172, Jan. 2003.



Costas G. Baslis (S'09–M'11) received the Dipl. Electr. Eng. and Ph.D. degrees from the Department of Electrical Engineering, Aristotle University of Thessaloniki, Thessaloniki, Greece, in 2003 and 2010, respectively.

His research interests are in power system planning and economics.



Anastasios G. Bakirtzis (S'77–M'79–SM'95) received the Dipl. Eng. Degree from the Department of Electrical Engineering, National Technical University, Athens, Greece, in 1979 and the M.S.E.E. and Ph.D. degrees from Georgia Institute of Technology, Atlanta, in 1981 and 1984, respectively.

Since 1986, he has been with the Electrical Engineering Department, Aristotle University of Thessaloniki, Thessaloniki, Greece, where he is currently Professor and Director of the Power Systems Laboratory. His research interests are in power system operation, planning, and economics.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 22:18:07 UTC from IEEE Xplore. Restrictions apply.